

Physics Packet: Forces on a Ramp

Box on a Level Surface

Consider a box resting on a frictionless horizontal surface, as shown in Figure .



Concept Questions

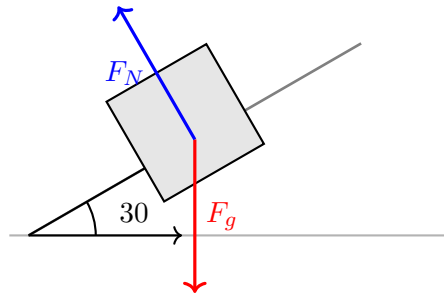
1. What do we know about the direction of the normal force F_N in this situation?

2. What do we know about the direction of the gravitational force F_g ?

3. If the box is at rest and the surface is frictionless, how do the magnitudes of F_N and F_g compare?

Box on an Inclined Plane (Ramp)

Now the box is placed on a ramp that makes an angle (30°), θ with the horizontal, as shown in the image below. The surface is still frictionless.



Observation Questions

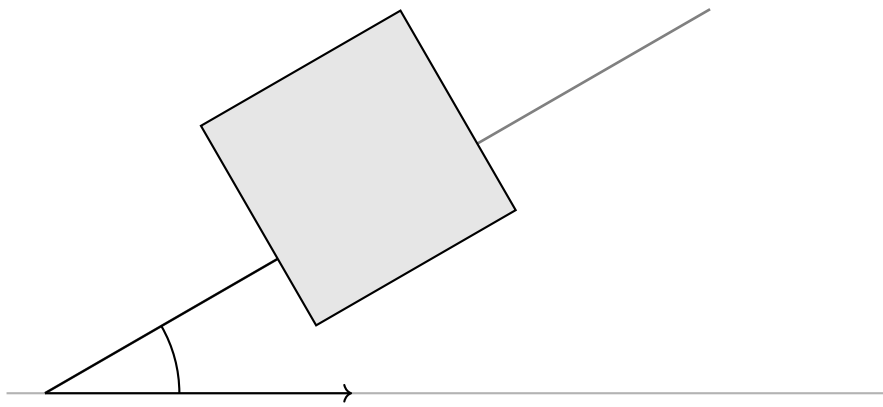
1. List out a few differences you see on the graph compared to the free body diagram on a horizontal surface:

2. In the ramp situation:

- (a) Why is the normal force F_N pointing up and tilted to the right?

- (b) Why is the gravitational force F_g still pointing straight down?

Breaking the Weight into Components on the Ramp



Practice Problems

Problem 1.

A 5.0 kg box rests on a frictionless ramp that makes an angle of 30° with the horizontal.

1. Draw a free body diagram for the box on the ramp.
2. Calculate the gravitational force F_g of the box.
3. Calculate F_{gx} . Show your work.
4. Calculate F_{gy} . Show your work.
5. If the box does not accelerate off or through the surface of the ramp, what is the value of the normal force F_N ?

Problem 2.

A 8.0 kg box rests on a frictionless ramp that makes an angle of 45° with the horizontal.

1. Draw a free body diagram for the box on the ramp.
2. Calculate the gravitational force F_g of the box.
3. Calculate F_{gx} . Show your work.
4. Calculate F_{gy} . Show your work.
5. If the box is not accelerating perpendicular to the ramp, what is the value of the normal force F_N ?

Problem 3. Challenge

A box rests on a frictionless ramp. Suppose F_{gx} is 96.3 N and F_{gy} is 67.5 N .

1. Draw a free body diagram for the box on the ramp.

2. Calculate the gravitational force F_g of the box.

3. Calculate m . Show your work.

4. Calculate F_{gx} . Show your work.

5. Calculate F_{gy} . Show your work.

6. Calculate F_N . Show your work.

7. Calculate the angle of the ramp.